GCD, Bézout's and Gauss's theorems

Lecture and worked examples

Objectives

Find a gcd with Euclid's algorithm; obtain a Bézout pair; use Bézout's and Gauss's theorems; find an inverse modulo n; solve an equation $ax + by = c$ in integers.

Introductory activity

We have a 9-litre can, a 7-litre can, both without markings, and a water supply. We may fill a can, empty it, or pour one into the other until one is empty or the other is full.

Introductory activity (2)

1. Show that one can get exactly 2 litres in the large can.
2. If we pour in u times 9 litres altogether and throw away v times 7 litres, the quantity left is $9u - 7v$. Find natural numbers u and v such that $9u - 7v = 1$. Deduce how to get 1 litre.
3. The cans are replaced by cans of 6 and 9 litres. Show that every quantity of the form $6u + 9v$, with u and v integers, is a multiple of 3. Can one get 1 litre?

Introductory activity (3)

4. With cans of a and b litres, which whole quantities does it seem possible to get? State a conjecture involving the greatest common divisor of a and b.

GCD and Euclid's algorithm

Let a and b be two integers, not both zero. The set of their positive common divisors is finite and contains 1; its largest element is called the *greatest common divisor* of a and b, written $\gcd(a, b)$. We say that a and b are *coprime* if $\gcd(a, b) = 1$.

GCD and Euclid's algorithm

Let a and b be two integers with $b \neq 0$. For every integer k, the common divisors of a and b are those of b and $a - kb$. In particular, if r is the remainder of the division of a by $b \geq 1$, then $\gcd(a, b) = \gcd(b, r)$.

GCD and Euclid's algorithm

If $d \mid a$ and $d \mid b$, then $d \mid a - kb$. Conversely, if $d \mid b$ and $d \mid a - kb$, then $d \mid (a - kb) + kb = a$. The two sets of common divisors are equal, so their largest elements are too.

GCD and Euclid's algorithm

Replace (a, b) by (b, r) as long as the remainder r is not zero. The remainders form a strictly decreasing sequence of natural numbers, so the algorithm stops. The gcd is the last nonzero remainder. Example:

$$1{,}053 = 3 \times 325 + 78, \qquad 325 = 4 \times 78 + 13, \qquad 78 = 6 \times 13 + 0,$$

so $\gcd(1{,}053, 325) = 13$.

Bézout's theorem

Let a and b be two integers, not both zero. There are integers u and v such that $au + bv = \gcd(a, b)$. Moreover, every common divisor of a and b divides $\gcd(a, b)$.

Bézout's theorem

Let E be the set of positive integers of the form $au + bv$, with u, v integers. E is not empty (it contains $a^2 + b^2$). Let $d = au_0 + bv_0$ be its smallest element.

Divide a by d: $a = dq + r$ with $0 \leq r < d$. Then $r = a - q(au_0 + bv_0) = a(1 - qu_0) + b(-qv_0)$ is of the form $au + bv$. If it were positive, it would belong to E while being smaller than d: impossible. So $r = 0$ and $d \mid a$. Likewise $d \mid b$.

Bézout's theorem (2)

Finally every common divisor c of a and b divides $au_0 + bv_0 = d$, so $c \leq d$. Hence d is the greatest common divisor: $d = \gcd(a, b)$.

Bézout's theorem

Two integers a and b are coprime if and only if there are integers u and v such that $au + bv = 1$.

Bézout's theorem

If $\gcd(a, b) = 1$, the previous theorem provides u and v. Conversely, if $au + bv = 1$, every positive common divisor of a and b divides 1, hence equals 1.

Bézout's theorem

With the previous example:
$13 = 325 - 4 \times 78 = 325 - 4(1{,}053 - 3 \times 325) = 13 \times 325 - 4 \times 1{,}053$. A suitable pair for $(1{,}053, 325)$ is $(-4, 13)$.

Bézout's theorem

The following function returns (d, u, v) with $d = \gcd(a, b) = au + bv$, for natural numbers a and b not both zero. lstlisting def euclide_etendu(a, b): u0, v0, u1, v1 = 1, 0, 0, 1 while b != 0: q, r = a // b, a % b a, b = b, r u0, u1 = u1, u0 - q * u1 v0, v1 = v1, v0 - q * v1 return a, u0, v0 lstlisting At every step, the two current remainders can be written $au_0 + bv_0$ and $au_1 + bv_1$ in terms of the initial values of a and b.

Gauss's theorem

Let a, b, c be three integers. If $a \mid bc$ and a and b are coprime, then $a \mid c$.

Gauss's theorem

By Bézout's theorem, there are u and v such that $au + bv = 1$. Multiplying by c: $c = acu + bcv$. Now $a \mid acu$ and $a \mid bc$, so a divides $acu + bcv = c$.

Gauss's theorem

If a and b are coprime and both divide n, then ab divides n.

Gauss's theorem

$n = ak$. Since $b \mid ak$ and $\gcd(a, b) = 1$, Gauss gives $b \mid k$: $k = bk'$ and $n = abk'$.

Gauss's theorem

Let $n \geq 2$ and a an integer. There is an integer u such that $au \equiv 1 \pmod{n}$ if and only if $\gcd(a, n) = 1$. Such a u is an inverse of a modulo n; it allows us to solve $ax \equiv b \pmod{n}$: $x \equiv ub \pmod{n}$.

Gauss's theorem

$au \equiv 1 \pmod{n}$ is equivalent to the existence of an integer v such that $au + nv = 1$: this is Bézout's theorem.

Gauss's theorem

1. It has solutions if and only if $\gcd(a, b)$ divides c; we then divide by this gcd.
2. Find a particular solution (x_0, y_0).
3. By subtraction, $a(x - x_0) = -b(y - y_0)$; with a and b coprime, Gauss gives $x - x_0 = bk$, then $y - y_0 = -ak$.
4. Check that the pairs $(x_0 + bk, y_0 - ak)$, $k \in \mathbb{Z}$, are indeed solutions.

Example: $7x + 5y = 1$ has the particular solution $(-2, 3)$; its solutions are the pairs $(-2 + 5k, 3 - 7k)$, $k \in \mathbb{Z}$.